JP Morgan ML Take-Home Assignment

This notebook covers both requested deliverables:

1. A classification model to predict whether income is above or below $50k.
2. A customer segmentation model for marketing use.

1. Imports and Setup

In [ ]:
```
# ============================================================
# 1. Setup
# ============================================================

import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder, StandardScaler

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier

from sklearn.metrics import (
    roc_auc_score,
    average_precision_score,
    precision_recall_curve,
    roc_curve,
    confusion_matrix,
    ConfusionMatrixDisplay
)

RANDOM_STATE = 42
TEST_SIZE = 0.2
```

2. Load the Data

The dataset comes with a separate file for column names, so we first reconstruct a clean DataFrame.

In [2]:
```
# My first goal here is simply to reconstruct a readable dataframe from the raw
# I do this explicitly instead of relying on default headers because the assignm
# provides the column names separately and I wanted the setup to stay reproducib

# ============================================================
```

```python
# 2. Load data
# Replace the file path if needed
# ============================================================
COLUMN_FILE = 'census-bureau.columns'
DATA_FILE = 'census-bureau.data'

def load_census_data(column_path=COLUMN_FILE, data_path=DATA_FILE):
    columns = pd.read_csv(column_path, header=None).iloc[:, 0].str.strip().tolis
    df = pd.read_csv(data_path, header=None, names=columns)
    df.columns = [col.strip().lower().replace(' ', '_').replace("'", '') for col

    object_cols = df.select_dtypes(include='object').columns
    df[object_cols] = df[object_cols].apply(lambda col: col.str.strip())
    df = df.replace('?', np.nan)

    df['target'] = (df['label'].str.contains(r'50000\+', regex=True)).astype(int
    return df


df = load_census_data()
print(f'Shape: {df.shape}')
display(df.head())
```

```
Shape: (199523, 43)
```

	age	class_of_worker	detailed_industry_recode	detailed_occupation_recode	educatio
0	73	Not in universe	0	0	Hig schoo graduat
1	58	Self-employed-not incorporated	4	34	Som colleg but n degre
2	18	Not in universe	0	0	10t grad
3	9	Not in universe	0	0	Childre
4	10	Not in universe	0	0	Childre

5 rows × 43 columns

In [3]:
```python
# ============================================================
# 3. Create binary target
# The original label column is converted into a binary target:
# 1 = income > 50K, 0 = income <= 50K
# ============================================================

df = df.copy()
```

```python
# Adjust this line if your label text is slightly different
df["target"] = (df["label"].astype(str).str.contains("50000\\+")).astype(int)

print(df["target"].value_counts(dropna=False))
print(df["target"].value_counts(normalize=True))
```

```
target
0    187141
1     12382
Name: count, dtype: int64
target
0    0.937942
1    0.062058
Name: proportion, dtype: float64
```

In [4]:

```python
# ============================================================
# 4. Basic schema check
# ============================================================

print("Shape:", df.shape)
print("\nColumn types:")
print(df.dtypes.value_counts())

missing_summary = pd.DataFrame({
    "dtype": df.dtypes.astype(str),
    "missing_count": df.isna().sum(),
    "missing_pct": df.isna().mean().round(4),
    "n_unique": df.nunique(dropna=True)
}).sort_values(["missing_count", "n_unique"], ascending=[False, False])

display(missing_summary.head(20))
```

```
Shape: (199523, 43)

Column types:
object     29
int64      13
float64     1
Name: count, dtype: int64
```

	dtype	missing_count	missing_pct	n_unique
migration_code-change_in_msa	object	99696	0.4997	9
migration_code-move_within_reg	object	99696	0.4997	9
migration_code-change_in_reg	object	99696	0.4997	8
migration_prev_res_in_sunbelt	object	99696	0.4997	3
country_of_birth_father	object	6713	0.0336	42
country_of_birth_mother	object	6119	0.0307	42
country_of_birth_self	object	3393	0.0170	42
hispanic_origin	object	874	0.0044	9
state_of_previous_residence	object	708	0.0035	50
weight	float64	0	0.0000	99800
dividends_from_stocks	int64	0	0.0000	1478
wage_per_hour	int64	0	0.0000	1240
capital_gains	int64	0	0.0000	132
capital_losses	int64	0	0.0000	113
age	int64	0	0.0000	91
weeks_worked_in_year	int64	0	0.0000	53
detailed_industry_recode	int64	0	0.0000	52
detailed_occupation_recode	int64	0	0.0000	47
detailed_household_and_family_stat	object	0	0.0000	38
major_industry_code	object	0	0.0000	24

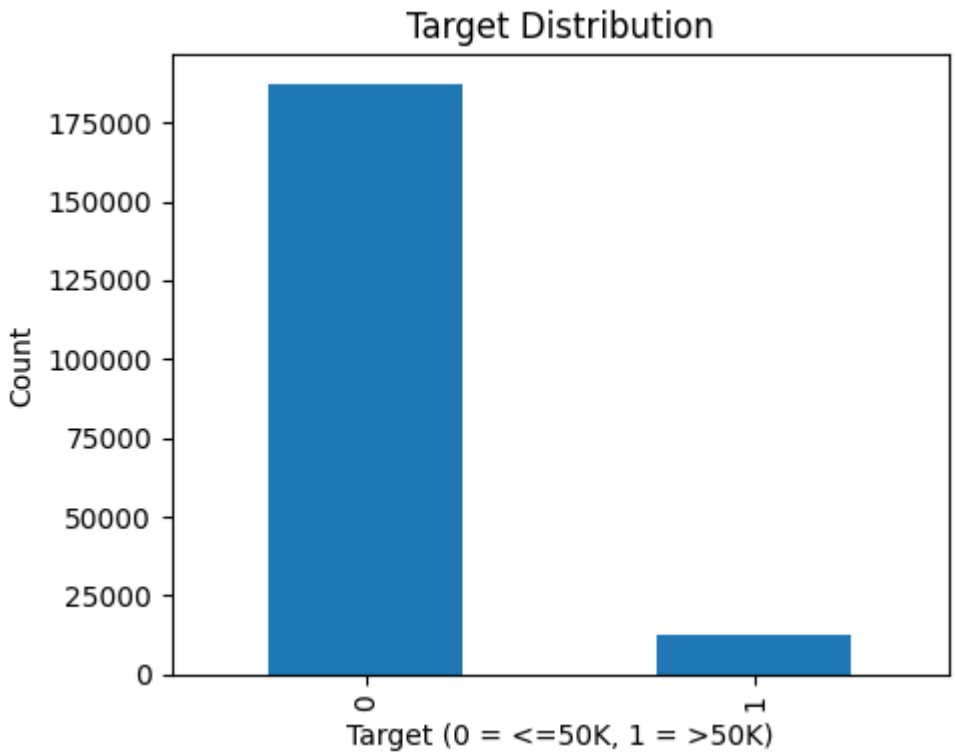
In [5]:
```
# ============================================================
# 5. Keep a clean working copy
# ============================================================

df_model = df.copy()
```

3. Quick Data Audit

In [6]:
```
# ============================================================
# 6. Target distribution
# The positive class is relatively rare, so PR-AUC and F1
# will be especially important evaluation metrics.
# ============================================================

target_dist = df_model["target"].value_counts().sort_index()
target_prop = df_model["target"].value_counts(normalize=True).sort_index()

print("Target counts:")
print(target_dist)
print("\nTarget proportions:")
print(target_prop)
```

```
plt.figure(figsize=(5, 4))
target_dist.plot(kind="bar")
plt.title("Target Distribution")
plt.xlabel("Target (0 = <=50K, 1 = >50K)")
plt.ylabel("Count")
plt.tight_layout()
plt.show()
```

```
Target counts:
target
0    187141
1     12382
Name: count, dtype: int64

Target proportions:
target
0    0.937942
1    0.062058
Name: proportion, dtype: float64
```



In [7]:
```
# I use this quick summary to separate true missingness from fields that simply
# At this stage I am not fixing anything yet — I just want a high-level sense of
# the data may need more careful treatment later in preprocessing.

# ============================================================
# 7. Missingness overview
# ============================================================

missing_summary = pd.DataFrame({
    "dtype": df_model.dtypes.astype(str),
    "missing_count": df_model.isna().sum(),
    "missing_pct": df_model.isna().mean(),
    "n_unique": df_model.nunique(dropna=True)
}).sort_values(["missing_count", "missing_pct"], ascending=[False, False])

display(missing_summary.head(20))
```

	dtype	missing_count	missing_pct	n_unique
migration_code-change_in_msa	object	99696	0.499672	9
migration_code-change_in_reg	object	99696	0.499672	8
migration_code-move_within_reg	object	99696	0.499672	9
migration_prev_res_in_sunbelt	object	99696	0.499672	3
country_of_birth_father	object	6713	0.033645	42
country_of_birth_mother	object	6119	0.030668	42
country_of_birth_self	object	3393	0.017006	42
hispanic_origin	object	874	0.004380	9
state_of_previous_residence	object	708	0.003548	50
age	int64	0	0.000000	91
class_of_worker	object	0	0.000000	9
detailed_industry_recode	int64	0	0.000000	52
detailed_occupation_recode	int64	0	0.000000	47
education	object	0	0.000000	17
wage_per_hour	int64	0	0.000000	1240
enroll_in_edu_inst_last_wk	object	0	0.000000	3
marital_stat	object	0	0.000000	7
major_industry_code	object	0	0.000000	24
major_occupation_code	object	0	0.000000	15
race	object	0	0.000000	5

In [8]:

```
# A lot of the categorical columns use placeholder values like "Not in universe"
# I did not want to blindly treat those as standard nulls, so I inspect them sep
# For this dataset, those placeholders often carry real meaning about eligibilit

# ============================================================
# 8. Inspect categorical placeholders
# Many census-like fields contain values such as
# 'Not in universe' that are not ordinary missing values.
# ============================================================

obj_cols = df_model.select_dtypes(include="object").columns.tolist()

suspicious_tokens = [
    "not in universe", "unknown", "?", "not known",
    "other", "none", "missing", "not applicable", "n/a"
]

rows = []
for col in obj_cols:
    vc = df_model[col].astype(str).str.strip().str.lower().value_counts(dropna=F
    suspicious_found = [k for k in vc.index if any(tok in k for tok in suspiciou
    rows.append({
```

```
        "column": col,
        "n_unique": df_model[col].nunique(dropna=True),
        "top_values": dict(vc.head(8)),
        "suspicious_values_found": suspicious_found[:10]
    })

obj_summary = pd.DataFrame(rows).sort_values("n_unique", ascending=False)
display(obj_summary)
```

	column	n_unique	top_values	suspicious_valu
14	state_of_previous_residence	50	{'not in universe': 183750, 'california': 1714...	[not in
24	country_of_birth_mother	42	{'united-states': 160479, 'mexico': 9781, 'nan...	
23	country_of_birth_father	42	{'united-states': 159163, 'mexico': 10008, 'na...	
25	country_of_birth_self	42	{'united-states': 176989, 'mexico': 5767, 'nan...	
15	detailed_household_and_family_stat	38	{'householder': 53248, 'child <18 never marr n...	[other rel 18+ not in subfa
4	major_industry_code	24	{'not in universe or children': 100684, 'retai...	[not in u child pr
1	education	17	{'high school graduate': 48407, 'children': 47...	
5	major_occupation_code	15	{'not in universe': 100684, 'adm support inclu...	[not in unive
19	migration_code-move_within_reg	9	{'nan': 99696, 'nonmover': 82538, 'same county...	[not in
7	hispanic_origin	9	{'all other': 171907, 'mexican-american': 8079...	[all other, othe
17	migration_code-change_in_msa	9	{'nan': 99696, 'nonmover': 82538, 'msa to msa'...	[not in
0	class_of_worker	9	{'not in universe': 100245,	[not in

	column	n_unique	top_values	suspicious_valu
			'private': 72028, ...	
11	full_or_part_time_employment_stat	8	{'children or armed forces': 123769, 'full-tim...	
16	detailed_household_summary_in_household	8	{'householder': 75475, 'child under 18 never m...	[other hou
18	migration_code-change_in_reg	8	{'nan': 99696, 'nonmover': 82538, 'same county...	[not in
3	marital_stat	7	{'never married': 86485, 'married-civilian spo...	
13	region_of_previous_residence	6	{'not in universe': 183750, 'south': 4889, 'we...	[not in
10	reason_for_unemployment	6	{'not in universe': 193453, 'other job loser':...	[not in unive
12	tax_filer_stat	6	{'nonfiler': 75094, 'joint both under 65': 673...	
26	citizenship	5	{'native- born in the united states': 176992, ...	
6	race	5	{'white': 167365, 'black': 20415, 'asian or pa...	
22	family_members_under_18	5	{'not in universe': 144232, 'both parents pres...	[not in univers onl
20	live_in_this_house_1_year_ago	3	{'not in universe under 1 year old': 101212, '...	[not in univers
2	enroll_in_edu_inst_last_wk	3	{'not in universe':	[not in

	column	n_unique	top_values	suspicious_valu
			186943, 'high school': 689...	
9	member_of_a_labor_union	3	{'not in universe': 180459, 'no': 16034, 'yes'...	[not in
27	fill_inc_questionnaire_for_veterans_admin	3	{'not in universe': 197539, 'no': 1593, 'yes':...	[not in
21	migration_prev_res_in_sunbelt	3	{'nan': 99696, 'not in universe': 84054, 'no':...	[not in
8	sex	2	{'female': 103984, 'male': 95539}	
28	label	2	{'- 50000.': 187141, '50000+.': 12382}	

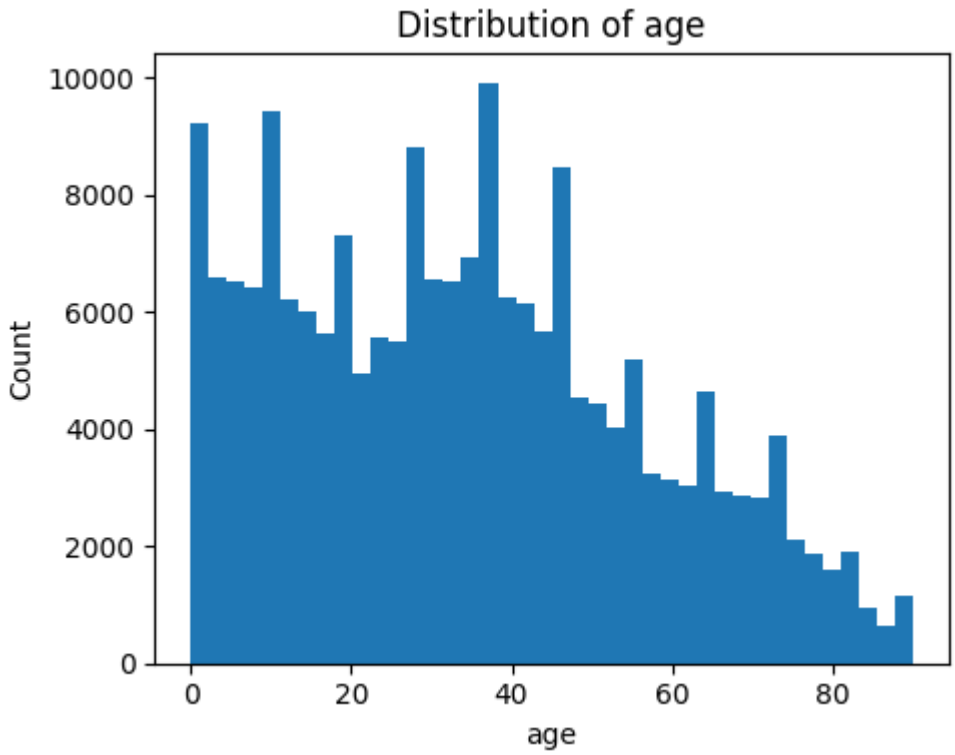
In [9]:

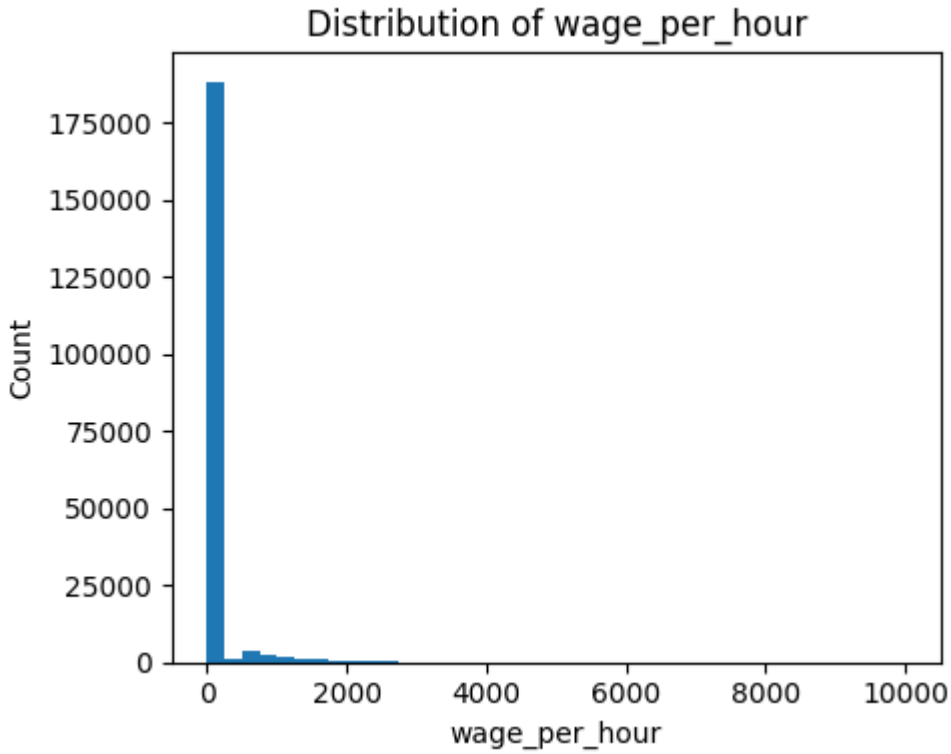
```
# ============================================================
# 9. Numeric profile
# Several numeric-looking fields are actually coded categories.
# ============================================================

num_cols = df_model.select_dtypes(include=["int64", "float64"]).columns.tolist()

num_profile = pd.DataFrame({
    "column": num_cols,
    "n_unique": [df_model[c].nunique(dropna=True) for c in num_cols],
    "min": [df_model[c].min() for c in num_cols],
    "max": [df_model[c].max() for c in num_cols]
}).sort_values("n_unique")

display(num_profile)
```

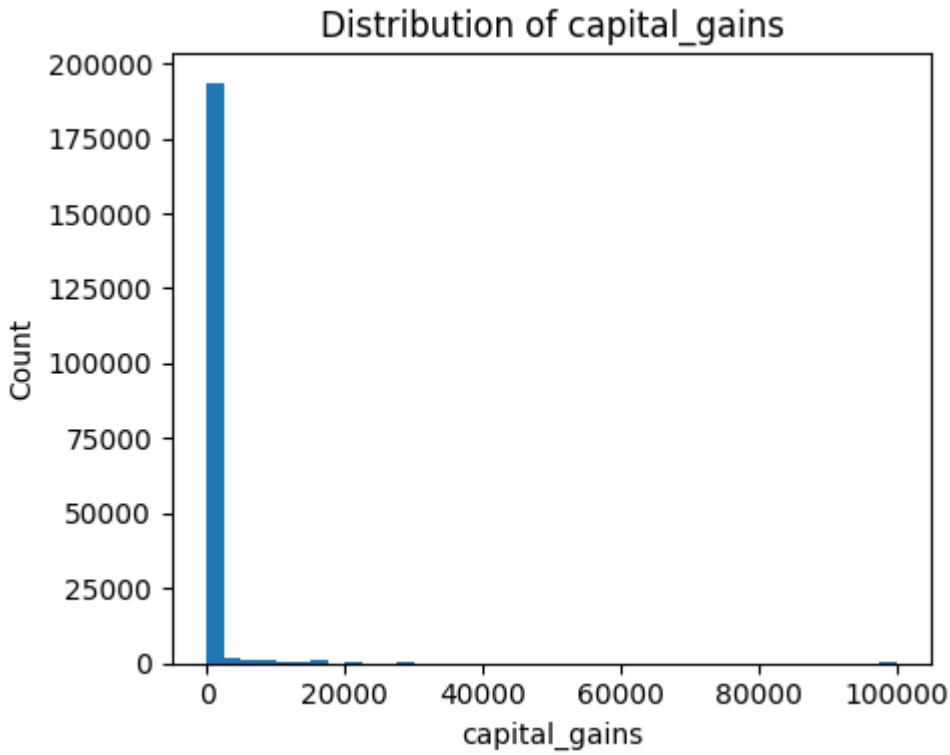
	column	n_unique	min	max
12	year	2	94.00	95.0
13	target	2	0.00	1.0
10	veterans_benefits	3	0.00	2.0
9	own_business_or_self_employed	3	0.00	2.0
8	num_persons_worked_for_employer	7	0.00	6.0
2	detailed_occupation_recode	47	0.00	46.0
1	detailed_industry_recode	52	0.00	51.0
11	weeks_worked_in_year	53	0.00	52.0
0	age	91	0.00	90.0
5	capital_losses	113	0.00	4608.0
4	capital_gains	132	0.00	99999.0
3	wage_per_hour	1240	0.00	9999.0
6	dividends_from_stocks	1478	0.00	99999.0
7	weight	99800	37.87	18656.3

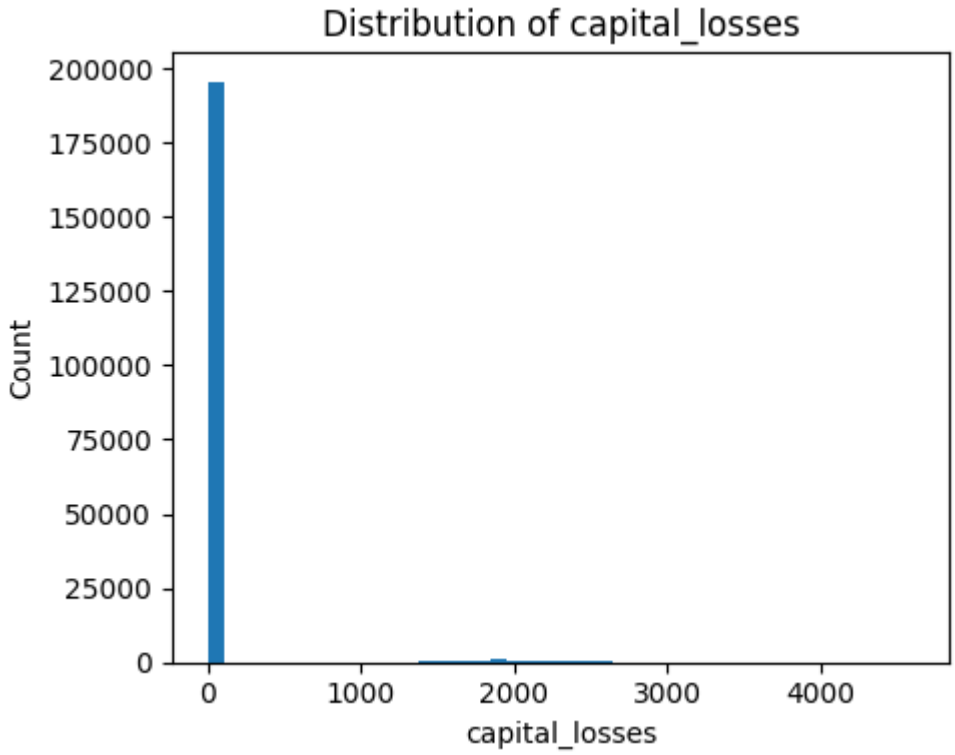
In [10]:

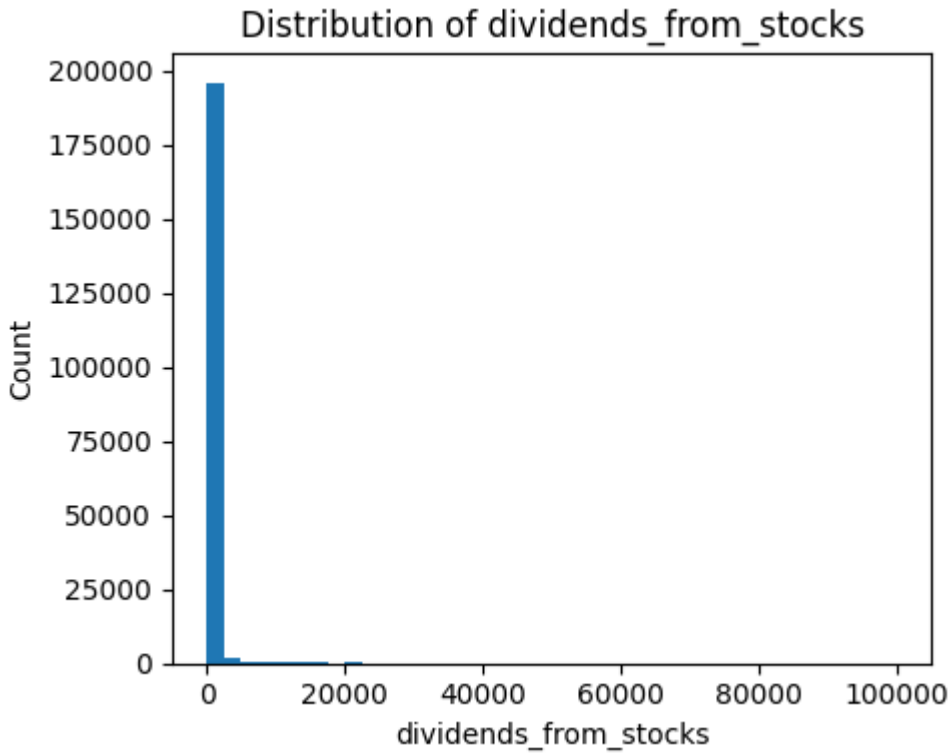
```
# ============================================================
# 10. A few quick numeric plots
# Keep the EDA concise and focused on variables likely related
# to income.
# ============================================================

eda_numeric_cols = [
    "age",
    "wage_per_hour",
    "capital_gains",
    "capital_losses",
    "dividends_from_stocks",
    "weeks_worked_in_year"
]

for col in eda_numeric_cols:
    plt.figure(figsize=(5, 4))
    plt.hist(df_model[col], bins=40)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Count")
    plt.tight_layout()
    plt.show()
```

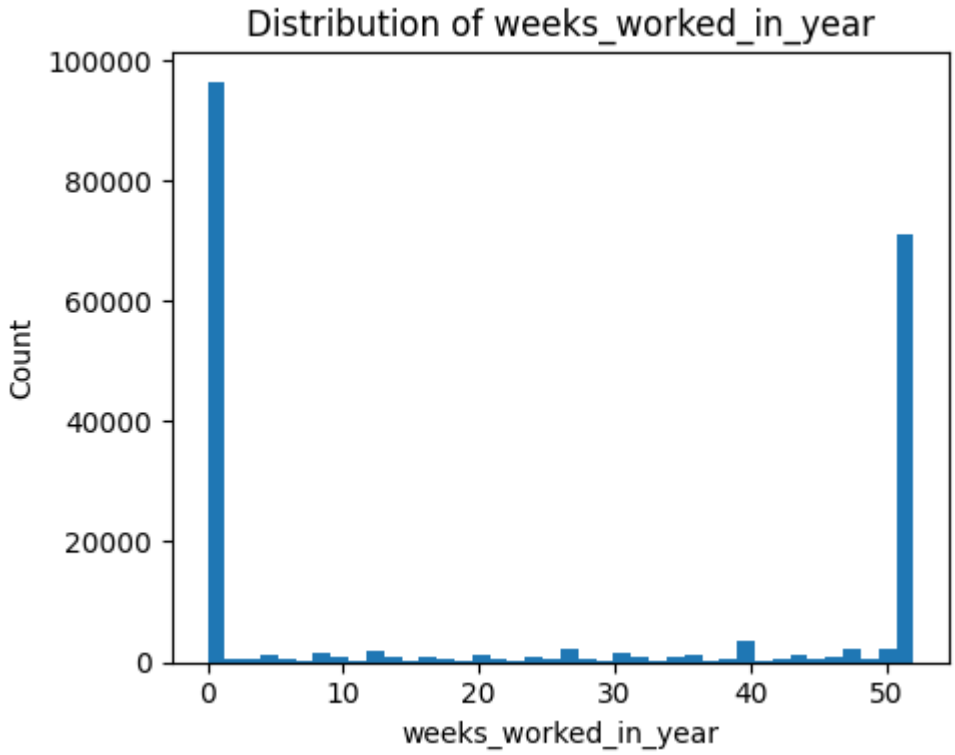

Distribution of age
Count
age
0
2000
4000
6000
8000
10000
0
20
40
60
80


Distribution of wage_per_hour
Count
wage_per_hour
0
25000
50000
75000
100000
125000
150000
175000
0
2000
4000
6000
8000
10000


Distribution of capital_gains
Count
capital_gains
200000
175000
150000
125000
100000
75000
50000
25000
0
0
20000
40000
60000
80000
100000


Distribution of capital_losses
Count
capital_losses
200000
175000
150000
125000
100000
75000
50000
25000
0
0
1000
2000
3000
4000





In [11]:

```
# ============================================================
# 11. Compare positive-class rates for a few suspicious fields
# This helps identify whether missingness or special values
# contain information.
# ============================================================

migration_cols = [
    "migration_code-change_in_msa",
    "migration_code-move_within_reg",
    "migration_code-change_in_reg",
    "migration_prev_res_in_sunbelt"
]
```

```python
for col in migration_cols:
    print("=" * 80)
    print(col)
    print(df_model[col].value_counts(dropna=False).head(10))
    print("\nTarget rate by missingness:")
    print(df_model.groupby(df_model[col].isna())["target"].mean())
    print()
```

```
================================================================================
migration_code-change_in_msa
migration_code-change_in_msa
NaN                 99696
Nonmover            82538
MSA to MSA          10601
NonMSA to nonMSA     2811
Not in universe      1516
MSA to nonMSA         790
NonMSA to MSA         615
Abroad to MSA         453
Not identifiable      430
Abroad to nonMSA       73
Name: count, dtype: int64

Target rate by missingness:
migration_code-change_in_msa
False    0.058491
True     0.065630
Name: target, dtype: float64

================================================================================
migration_code-move_within_reg
migration_code-move_within_reg
NaN                             99696
Nonmover                        82538
Same county                      9812
Different county same state      2797
Not in universe                  1516
Different state in South          973
Different state in West           679
Different state in Midwest        551
Abroad                            530
Different state in Northeast      431
Name: count, dtype: int64

Target rate by missingness:
migration_code-move_within_reg
False    0.058491
True     0.065630
Name: target, dtype: float64

================================================================================
migration_code-change_in_reg
migration_code-change_in_reg
NaN                               99696
Nonmover                          82538
Same county                        9812
Different county same state        2797
Not in universe                    1516
Different region                   1178
Different state same division       991
Abroad                              530
Different division same region      465
Name: count, dtype: int64

Target rate by missingness:
migration_code-change_in_reg
False    0.058491
True     0.065630
```

```
Name: target, dtype: float64

================================================================================
migration_prev_res_in_sunbelt
migration_prev_res_in_sunbelt
NaN                99696
Not in universe    84054
No                  9987
Yes                 5786
Name: count, dtype: int64

Target rate by missingness:
migration_prev_res_in_sunbelt
False    0.058491
True     0.065630
Name: target, dtype: float64
```

4. Classification Modeling - Logistic Regression

Modeling choices

- I use a train/test split with stratification because the income classes are imbalanced.
- I keep the `weight` column available as a sample-weight feature for the classifier because the assignment notes that it reflects population representation.
- Categorical features are imputed and one-hot encoded.
- Numeric features are median-imputed and scaled.
- The main model is logistic regression because it is strong, fast, and easy to explain in a business-facing report.

How I approached the feature space

Before I started fitting models, I paused here to clean up the feature types. A lot of the raw columns come from census-style data, so some fields look numeric at first glance but are really coded categories. I wanted to make that explicit before building any preprocessing pipeline, since treating those columns incorrectly would hurt both model quality and interpretability.

In [12]:
```
# I made these feature-type decisions before building the pipelines so the rest
# would stay cleaner. In particular, I recast coded numeric fields as categorica
# the values represented labels rather than a meaningful numeric scale.

# ============================================================
# 12. Feature design decisions
# - Keep structured placeholder categories as valid levels
# - Treat migration NaN as an explicit category
# - Recast coded numeric columns as categorical
# - Use a separate sample weight variable, not a predictor
# ============================================================

df_work = df_model.copy()
```

```python
coded_numeric_cols = [
    "year",
    "veterans_benefits",
    "own_business_or_self_employed",
    "num_persons_worked_for_employer",
    "detailed_industry_recode",
    "detailed_occupation_recode"
]

continuous_numeric_cols = [
    "age",
    "wage_per_hour",
    "capital_gains",
    "capital_losses",
    "dividends_from_stocks",
    "weeks_worked_in_year"
]

migration_cols = [
    "migration_code-change_in_msa",
    "migration_code-move_within_reg",
    "migration_code-change_in_reg",
    "migration_prev_res_in_sunbelt"
]

# Explicitly encode structural missingness for migration variables
for col in migration_cols:
    df_work[col] = df_work[col].fillna("Missing")

# Recast coded numeric fields as categorical
for col in coded_numeric_cols:
    df_work[col] = df_work[col].astype(str)
```

In [13]:

```python
# ============================================================
# 12. Feature design decisions
# - Keep structured placeholder categories as valid levels
# - Treat migration NaN as an explicit category
# - Recast coded numeric columns as categorical
# - Use a separate sample weight variable, not a predictor
# ============================================================

df_work = df_model.copy()

coded_numeric_cols = [
    "year",
    "veterans_benefits",
    "own_business_or_self_employed",
    "num_persons_worked_for_employer",
    "detailed_industry_recode",
    "detailed_occupation_recode"
]

continuous_numeric_cols = [
    "age",
    "wage_per_hour",
    "capital_gains",
    "capital_losses",
    "dividends_from_stocks",
    "weeks_worked_in_year"
```

```
]

migration_cols = [
    "migration_code-change_in_msa",
    "migration_code-move_within_reg",
    "migration_code-change_in_reg",
    "migration_prev_res_in_sunbelt"
]

# Explicitly encode structural missingness for migration variables
for col in migration_cols:
    df_work[col] = df_work[col].fillna("Missing")

# Recast coded numeric fields as categorical
for col in coded_numeric_cols:
    df_work[col] = df_work[col].astype(str)
```

In [14]:

```
# From this point on, I switch into a clean modeling dataset.
# I keep the original label, the binary target, and the survey weight conceptual
# because they play different roles in fitting and evaluation.

# ============================================================
# 13. Separate X, y, and sample weights
# Weight is kept for optional weighted training or evaluation,
# but is not used as a predictor.
# ============================================================

X = df_work.drop(columns=["label", "target"]).copy()
y = df_work["target"].copy()

sample_weight = X["weight"].copy() if "weight" in X.columns else pd.Series(np.on
X = X.drop(columns=["weight"])
```

In [15]:

```
# I use one stratified split and keep it fixed across all models.
# That way, any performance difference I see later is much easier to attribute t
# itself rather than to a lucky or unlucky split.

# ============================================================
# 14. Train/test split
# ============================================================

X_train, X_test, y_train, y_test, w_train, w_test = train_test_split(
    X,
    y,
    sample_weight,
    test_size=TEST_SIZE,
    stratify=y,
    random_state=RANDOM_STATE
)

categorical_features = X_train.select_dtypes(include=["object"]).columns.tolist(
numeric_features = X_train.select_dtypes(include=["int64", "float64"]).columns.t

print("Number of categorical features:", len(categorical_features))
print("Categorical features:")
print(categorical_features)

print("\nNumeric features:")
print(numeric_features)
```

```
Number of categorical features: 34
Categorical features:
['class_of_worker', 'detailed_industry_recode', 'detailed_occupation_recode', 'ed
ucation', 'enroll_in_edu_inst_last_wk', 'marital_stat', 'major_industry_code', 'm
ajor_occupation_code', 'race', 'hispanic_origin', 'sex', 'member_of_a_labor_unio
n', 'reason_for_unemployment', 'full_or_part_time_employment_stat', 'tax_filer_st
at', 'region_of_previous_residence', 'state_of_previous_residence', 'detailed_hou
sehold_and_family_stat', 'detailed_household_summary_in_household', 'migration_co
de-change_in_msa', 'migration_code-change_in_reg', 'migration_code-move_within_re
g', 'live_in_this_house_1_year_ago', 'migration_prev_res_in_sunbelt', 'num_person
s_worked_for_employer', 'family_members_under_18', 'country_of_birth_father', 'co
untry_of_birth_mother', 'country_of_birth_self', 'citizenship', 'own_business_or_
self_employed', 'fill_inc_questionnaire_for_veterans_admin', 'veterans_benefits',
'year']

Numeric features:
['age', 'wage_per_hour', 'capital_gains', 'capital_losses', 'dividends_from_stock
s', 'weeks_worked_in_year']
```

In [16]:
```
# I keep a dedicated preprocessing function for linear models because they are m
# to scaling and to the exact encoding scheme. Wrapping this in a function makes
# for me to tweak one design choice, such as min_frequency, without rewriting ev

# ============================================================
# 15. Build preprocessors
# Logistic regression uses scaling for numeric features.
# Tree-based models do not need scaling.
# ============================================================

def build_preprocessor_for_linear(min_freq=0.01):
    numeric_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler())
    ])

    categorical_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="constant", fill_value="Missing")),
        ("onehot", OneHotEncoder(handle_unknown="ignore", min_frequency=min_freq
    ])

    return ColumnTransformer([
        ("num", numeric_pipeline, numeric_features),
        ("cat", categorical_pipeline, categorical_features)
    ])


def build_preprocessor_for_tree(min_freq=0.01):
    numeric_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="median"))
    ])

    categorical_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="constant", fill_value="Missing")),
        ("onehot", OneHotEncoder(handle_unknown="ignore", min_frequency=min_freq
    ])

    return ColumnTransformer([
        ("num", numeric_pipeline, numeric_features),
        ("cat", categorical_pipeline, categorical_features)
    ])
```

Why I wrapped the evaluation logic into a function

At this point, I knew I was going to compare several models under the same train/test split and the same reporting criteria. Instead of repeating the same scoring code over and over, I bundled the fitting, probability scoring, threshold search, and confusion-matrix extraction into one helper function. That made it much easier for me to compare models fairly and keep the notebook consistent.

In [17]:
```
# I bundled the repeated evaluation steps into one helper so I could compare mod
# This function fits the pipeline, scores probabilities, searches for a threshol
# and then returns the exact metrics I need for both notebook analysis and repor

# ============================================================
# 16. Evaluation helper
# Returns ROC-AUC, PR-AUC, best threshold by F1, and confusion matrix
```

```
# ============================================================

def evaluate_model(pipe, X_train, y_train, X_test, y_test, sample_weight_train=N
    if use_weight and sample_weight_train is not None:
        pipe.fit(X_train, y_train, model__sample_weight=sample_weight_train)
    else:
        pipe.fit(X_train, y_train)

    y_prob = pipe.predict_proba(X_test)[:, 1]

    roc_auc = roc_auc_score(y_test, y_prob)
    pr_auc = average_precision_score(y_test, y_prob)

    precisions, recalls, thresholds = precision_recall_curve(y_test, y_prob)
    f1_scores = 2 * (precisions[:-1] * recalls[:-1]) / (precisions[:-1] + recall

    best_idx = np.argmax(f1_scores)
    best_threshold = thresholds[best_idx]
    best_f1 = f1_scores[best_idx]
    best_precision = precisions[best_idx]
    best_recall = recalls[best_idx]

    y_pred = (y_prob >= best_threshold).astype(int)
    cm = confusion_matrix(y_test, y_pred)

    return {
        "roc_auc": roc_auc,
        "pr_auc": pr_auc,
        "best_threshold": best_threshold,
        "best_f1": best_f1,
        "precision": best_precision,
        "recall": best_recall,
        "tn": cm[0, 0],
        "fp": cm[0, 1],
        "fn": cm[1, 0],
        "tp": cm[1, 1],
        "y_prob": y_prob,
        "y_pred": y_pred,
        "confusion_matrix": cm
    }
```

In [18]:

```
# ============================================================
# 15. Build preprocessors
# Logistic regression uses scaling for numeric features.
# Tree-based models do not need scaling.
# ============================================================

def build_preprocessor_for_linear(min_freq=0.01):
    numeric_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler())
    ])

    categorical_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="constant", fill_value="Missing")),
        ("onehot", OneHotEncoder(handle_unknown="ignore", min_frequency=min_freq
    ])

    return ColumnTransformer([
        ("num", numeric_pipeline, numeric_features),
```

```
        ("cat", categorical_pipeline, categorical_features)
    ])


def build_preprocessor_for_tree(min_freq=0.01):
    numeric_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="median"))
    ])

    categorical_pipeline = Pipeline([
        ("imputer", SimpleImputer(strategy="constant", fill_value="Missing")),
        ("onehot", OneHotEncoder(handle_unknown="ignore", min_frequency=min_freq
    ])

    return ColumnTransformer([
        ("num", numeric_pipeline, numeric_features),
        ("cat", categorical_pipeline, categorical_features)
    ])
```

In [19]:

```
# ============================================================
# 17. Baseline: Logistic Regression
# ============================================================

log_pipe = Pipeline([
    ("preprocessor", build_preprocessor_for_linear(min_freq=0.01)),
    ("model", LogisticRegression(max_iter=2000, solver="lbfgs"))
])

log_results = evaluate_model(
    log_pipe,
    X_train, y_train,
    X_test, y_test,
    sample_weight_train=w_train,
    use_weight=False
)

pd.DataFrame([{
    "model": "LogisticRegression",
    "roc_auc": log_results["roc_auc"],
    "pr_auc": log_results["pr_auc"],
    "best_threshold": log_results["best_threshold"],
    "best_f1": log_results["best_f1"],
    "precision": log_results["precision"],
    "recall": log_results["recall"],
    "tn": log_results["tn"],
    "fp": log_results["fp"],
    "fn": log_results["fn"],
    "tp": log_results["tp"]
}])
```

Out[19]:

	model	roc_auc	pr_auc	best_threshold	best_f1	precision	rec
0	LogisticRegression	0.945792	0.612175	0.26916	0.579379	0.548538	0.6138

Baseline model comparison

I did not want to jump straight to XGBoost without a reference point, so I first compared three models that play different roles:

- **Logistic Regression** as a simple and explainable baseline
- **Random Forest** as a strong non-linear tree baseline
- **XGBoost** as the more flexible boosted-tree option

My goal here was not just to pick the best metric mechanically, but to see whether the extra model complexity was actually buying me a meaningful performance gain.

In [20]:
```
# I start with side-by-side baseline models before I commit to a final recommend
# This helps me avoid overclaiming that the boosted model is best unless it clea

# ============================================================
# 18. Compare multiple models
# ============================================================

model_results = []

# Logistic Regression
log_pipe = Pipeline([
    ("preprocessor", build_preprocessor_for_linear(min_freq=0.01)),
    ("model", LogisticRegression(max_iter=2000, solver="lbfgs"))
])

res_log = evaluate_model(log_pipe, X_train, y_train, X_test, y_test, use_weight=
model_results.append({
    "model": "LogisticRegression",
    "roc_auc": res_log["roc_auc"],
    "pr_auc": res_log["pr_auc"],
    "best_threshold": res_log["best_threshold"],
    "best_f1": res_log["best_f1"],
    "precision": res_log["precision"],
    "recall": res_log["recall"],
    "tn": res_log["tn"],
    "fp": res_log["fp"],
    "fn": res_log["fn"],
    "tp": res_log["tp"]
})

# Random Forest
rf_pipe = Pipeline([
    ("preprocessor", build_preprocessor_for_tree(min_freq=0.01)),
    ("model", RandomForestClassifier(
        n_estimators=300,
        max_depth=16,
        min_samples_split=10,
        min_samples_leaf=4,
        max_features="sqrt",
        random_state=RANDOM_STATE,
        n_jobs=-1
    ))
])

res_rf = evaluate_model(rf_pipe, X_train, y_train, X_test, y_test, use_weight=Fa
model_results.append({
    "model": "RandomForest",
    "roc_auc": res_rf["roc_auc"],
```

```
    "pr_auc": res_rf["pr_auc"],
    "best_threshold": res_rf["best_threshold"],
    "best_f1": res_rf["best_f1"],
    "precision": res_rf["precision"],
    "recall": res_rf["recall"],
    "tn": res_rf["tn"],
    "fp": res_rf["fp"],
    "fn": res_rf["fn"],
    "tp": res_rf["tp"]
})

# XGBoost
xgb_pipe = Pipeline([
    ("preprocessor", build_preprocessor_for_tree(min_freq=0.01)),
    ("model", XGBClassifier(
        n_estimators=400,
        max_depth=5,
        learning_rate=0.05,
        subsample=0.8,
        colsample_bytree=0.8,
        min_child_weight=3,
        gamma=0,
        reg_alpha=0,
        reg_lambda=1,
        objective="binary:logistic",
        eval_metric="logloss",
        random_state=RANDOM_STATE,
        n_jobs=-1
    ))
])

res_xgb = evaluate_model(xgb_pipe, X_train, y_train, X_test, y_test, use_weight=
model_results.append({
    "model": "XGBoost",
    "roc_auc": res_xgb["roc_auc"],
    "pr_auc": res_xgb["pr_auc"],
    "best_threshold": res_xgb["best_threshold"],
    "best_f1": res_xgb["best_f1"],
    "precision": res_xgb["precision"],
    "recall": res_xgb["recall"],
    "tn": res_xgb["tn"],
    "fp": res_xgb["fp"],
    "fn": res_xgb["fn"],
    "tp": res_xgb["tp"]
})

model_results_df = pd.DataFrame(model_results).sort_values("best_f1", ascending=
model_results_df
```

Out[20]:

	model	roc_auc	pr_auc	best_threshold	best_f1	precision	rec
2	XGBoost	0.956083	0.695148	0.292965	0.630499	0.616275	0.6453
1	RandomForest	0.949795	0.654142	0.247540	0.602520	0.571325	0.6373
0	LogisticRegression	0.945792	0.612175	0.269160	0.579379	0.548538	0.6138

Narrow tuning after the baseline comparison

Once XGBoost emerged as the strongest candidate, I kept the follow-up tuning intentionally light. I was not trying to run a massive hyperparameter search in a take-home setting. Instead, I focused on a small design choice that mattered for sparse one-hot features: the `min_frequency` setting used in preprocessing. This let me check whether a slightly more permissive encoding improved recall and PR-AUC without turning the notebook into a tuning exercise.

In [21]:
```
# Once XGBoost looked strongest, I only did a narrow follow-up check here.
# I deliberately kept this tuning lightweight because I wanted the notebook to s
# not just a brute-force search over a huge parameter grid.

# ============================================================
# 19. Test how much rare-category grouping affects XGBoost
# ============================================================

xgb_minfreq_results = []

for mf in [0.01, 0.005]:
    pipe = Pipeline([
        ("preprocessor", build_preprocessor_for_tree(min_freq=mf)),
        ("model", XGBClassifier(
            n_estimators=400,
            max_depth=5,
            learning_rate=0.05,
            subsample=0.8,
            colsample_bytree=0.8,
            min_child_weight=3,
            gamma=0,
            reg_alpha=0,
            reg_lambda=1,
            objective="binary:logistic",
            eval_metric="logloss",
            random_state=RANDOM_STATE,
            n_jobs=-1
        ))
    ])

    res = evaluate_model(pipe, X_train, y_train, X_test, y_test, use_weight=Fals
    xgb_minfreq_results.append({
        "min_frequency": mf,
        "roc_auc": res["roc_auc"],
        "pr_auc": res["pr_auc"],
        "best_threshold": res["best_threshold"],
        "best_f1": res["best_f1"],
        "precision": res["precision"],
        "recall": res["recall"],
        "tn": res["tn"],
        "fp": res["fp"],
        "fn": res["fn"],
        "tp": res["tp"]
    })

xgb_minfreq_df = pd.DataFrame(xgb_minfreq_results).sort_values("best_f1", ascend
xgb_minfreq_df
```

Out[21]:

	min_frequency	roc_auc	pr_auc	best_threshold	best_f1	precision	recall
1	0.005	0.955962	0.694819	0.286896	0.633509	0.612599	0.655897
0	0.010	0.956083	0.695148	0.292965	0.630499	0.616275	0.645396

In [22]:

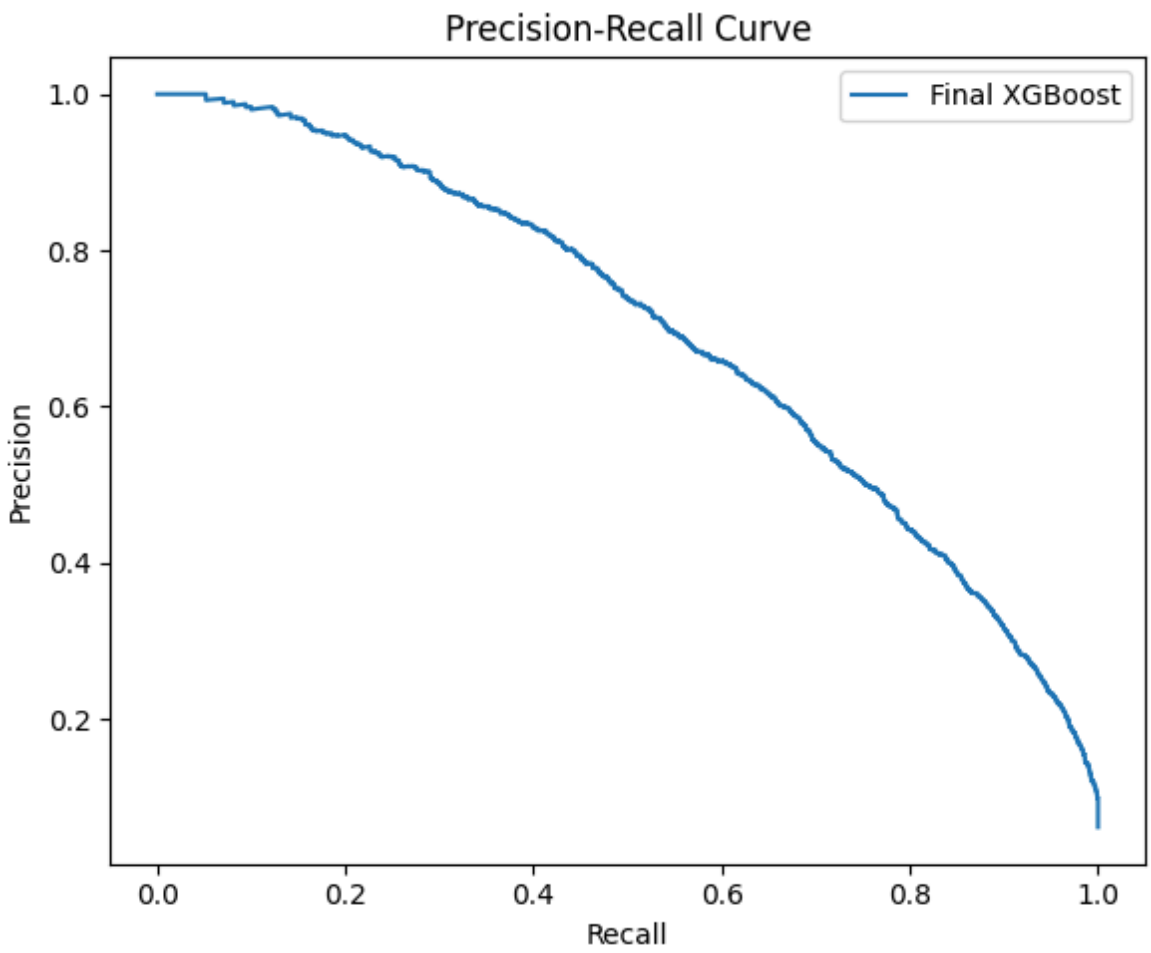
```
# This is the final training block I use to lock in the recommended model.
# After choosing the preprocessing and thresholding strategy, I refit the pipeli
# and store the outputs that I later reuse in the report figures.

# ============================================================
# 20. Final model
# Use the XGBoost configuration that gave the best F1.
# ============================================================

final_pipe = Pipeline([
    ("preprocessor", build_preprocessor_for_tree(min_freq=0.005)),
    ("model", XGBClassifier(
        n_estimators=400,
        max_depth=5,
        learning_rate=0.05,
        subsample=0.8,
        colsample_bytree=0.8,
        min_child_weight=3,
        gamma=0,
        reg_alpha=0,
        reg_lambda=1,
        objective="binary:logistic",
        eval_metric="logloss",
        random_state=RANDOM_STATE,
        n_jobs=-1
    ))
])

final_results = evaluate_model(
    final_pipe,
    X_train, y_train,
    X_test, y_test,
    use_weight=False
)

pd.DataFrame([{
    "model": "Final_XGBoost",
    "roc_auc": final_results["roc_auc"],
    "pr_auc": final_results["pr_auc"],
    "best_threshold": final_results["best_threshold"],
    "best_f1": final_results["best_f1"],
    "precision": final_results["precision"],
    "recall": final_results["recall"],
    "tn": final_results["tn"],
    "fp": final_results["fp"],
    "fn": final_results["fn"],
    "tp": final_results["tp"]
}])
```

Out[22]:

	model	roc_auc	pr_auc	best_threshold	best_f1	precision	recall
0	Final_XGBoost	0.955962	0.694819	0.286896	0.633509	0.612599	0.655897

In [23]:

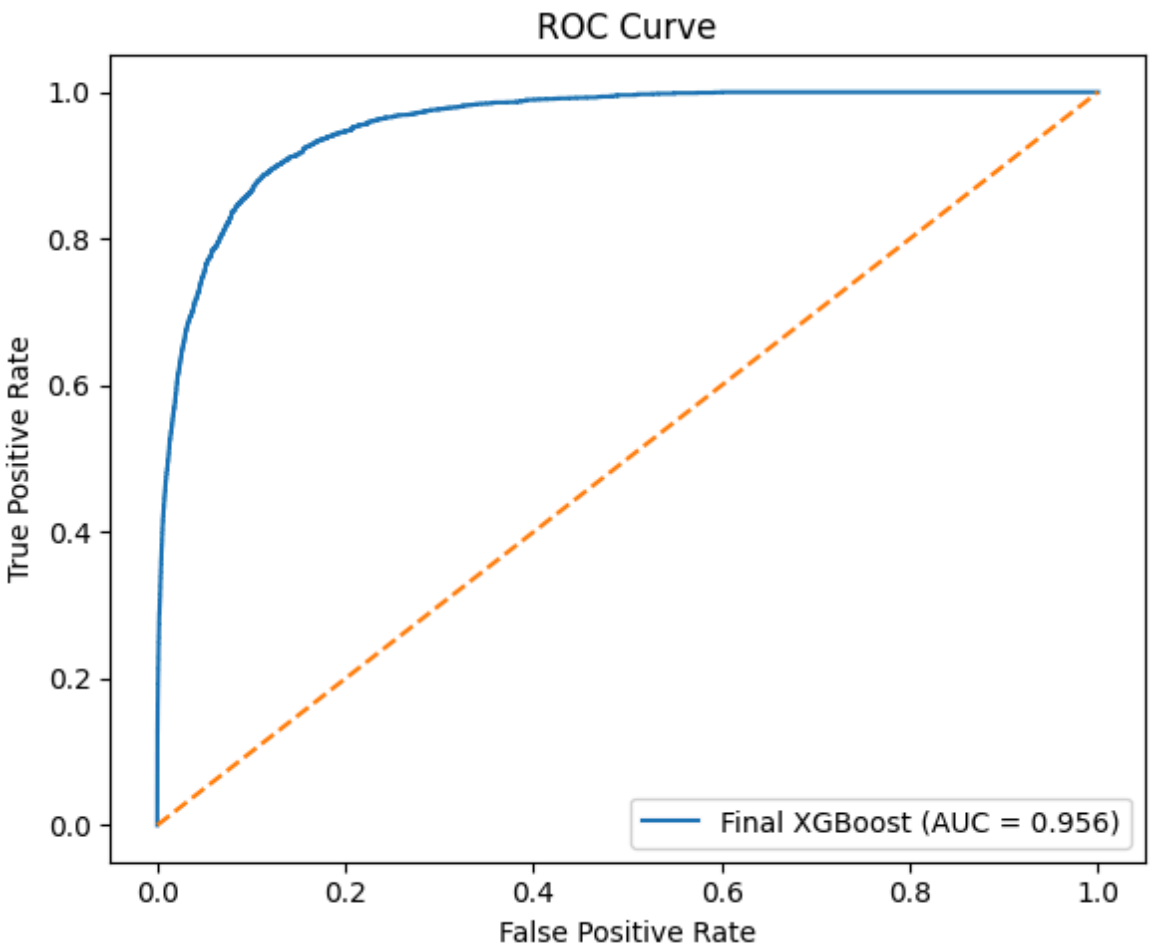
```
# ============================================================
# 21. Precision-recall curve for the final model
# ============================================================

y_prob_final = final_results["y_prob"]

precisions, recalls, _ = precision_recall_curve(y_test, y_prob_final)

plt.figure(figsize=(6, 5))
plt.plot(recalls, precisions, label="Final XGBoost")
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve")
plt.legend()
plt.tight_layout()
plt.show()
```



In [24]:

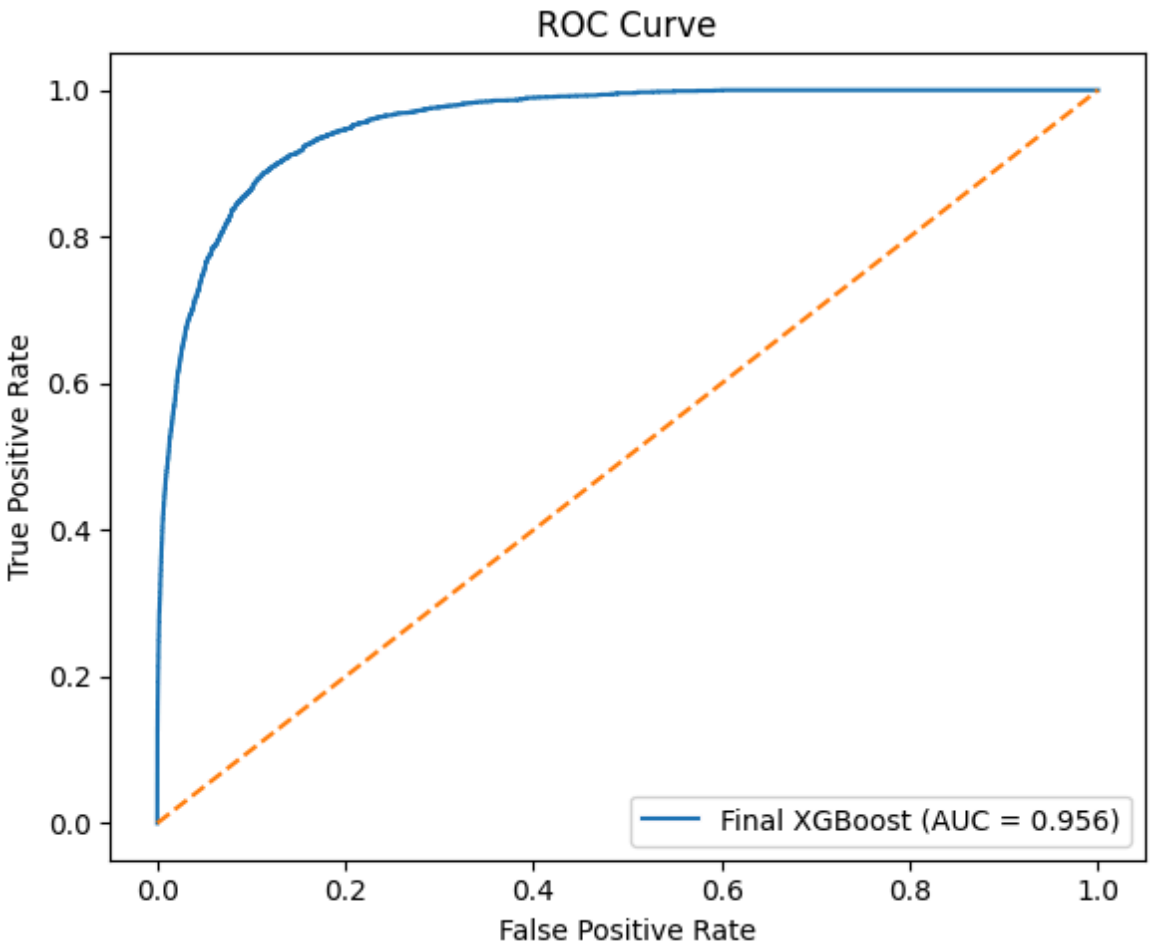
```
# ============================================================
# 22. ROC curve for the final model
# ============================================================

fpr, tpr, _ = roc_curve(y_test, y_prob_final)

plt.figure(figsize=(6, 5))
```

```
plt.plot(fpr, tpr, label=f"Final XGBoost (AUC = {final_results['roc_auc']:.3f})"
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.tight_layout()
plt.show()
```



In [25]:

```
# ============================================================
# 22. ROC curve for the final model
# ============================================================

fpr, tpr, _ = roc_curve(y_test, y_prob_final)

plt.figure(figsize=(6, 5))
plt.plot(fpr, tpr, label=f"Final XGBoost (AUC = {final_results['roc_auc']:.3f})"
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.tight_layout()
plt.show()
```



In [26]:
```
# I combine the key outputs into one summary table so I can reuse the same resul
# This cell is mostly for reporting convenience rather than core modeling logic.

# ============================================================
# 24. Final summary table
# ============================================================

summary_df = pd.concat([
    model_results_df.assign(section="Model comparison"),
    xgb_minfreq_df.assign(section="XGBoost min_frequency test")
], ignore_index=True)

summary_df
```

Out[26]:

	model	roc_auc	pr_auc	best_threshold	best_f1	precision	rec
0	XGBoost	0.956083	0.695148	0.292965	0.630499	0.616275	0.6453
1	RandomForest	0.949795	0.654142	0.247540	0.602520	0.571325	0.6373
2	LogisticRegression	0.945792	0.612175	0.269160	0.579379	0.548538	0.6138
3	NaN	0.955962	0.694819	0.286896	0.633509	0.612599	0.6558
4	NaN	0.956083	0.695148	0.292965	0.630499	0.616275	0.6453

Turning the modeling results into report-ready outputs

From here on, the remaining cells are mostly about packaging the results into tables and visuals that I could use in the written report. A few of these cells overlap with one another because I tried more than one way to present the same result. I kept those iterations on purpose so the reader can still see how I refined the final reporting view.

In [27]:

```
# I re-import plotting libraries here on purpose so this reporting section is se
# Some of the cells below overlap with one another because I tried several layou
# how I wanted to present the final results in the write-up.

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

viz_df = summary_df.copy()

# Fill model names for the min_frequency experiment rows
mask = viz_df["model"].isna() & (viz_df["section"] == "XGBoost min_frequency tes
viz_df.loc[mask, "model"] = "XGBoost"

# Create a display label for plotting
viz_df["display_name"] = viz_df["model"].astype(str)

mf_mask = viz_df["section"] == "XGBoost min_frequency test"
viz_df.loc[mf_mask, "display_name"] = (
    viz_df.loc[mf_mask, "model"] + " (min_freq=" + viz_df.loc[mf_mask, "min_freq
)

display(viz_df)
```

	model	roc_auc	pr_auc	best_threshold	best_f1	precision	recal
0	XGBoost	0.956083	0.695148	0.292965	0.630499	0.616275	0.645396
1	RandomForest	0.949795	0.654142	0.247540	0.602520	0.571325	0.637318
2	LogisticRegression	0.945792	0.612175	0.269160	0.579379	0.548538	0.613893
3	XGBoost	0.955962	0.694819	0.286896	0.633509	0.612599	0.655897
4	XGBoost	0.956083	0.695148	0.292965	0.630499	0.616275	0.645396

In [28]:

```
# This cell is very similar to the previous plotting block.
# I kept it because I used it while iterating on figure layout and captions, but
# I only needed one final version for the report.

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

viz_df = summary_df.copy()

# Fill model names for the min_frequency experiment rows
mask = viz_df["model"].isna() & (viz_df["section"] == "XGBoost min_frequency tes
viz_df.loc[mask, "model"] = "XGBoost"

# Create a display label for plotting
viz_df["display_name"] = viz_df["model"].astype(str)

mf_mask = viz_df["section"] == "XGBoost min_frequency test"
viz_df.loc[mf_mask, "display_name"] = (
    viz_df.loc[mf_mask, "model"] + " (min_freq=" + viz_df.loc[mf_mask, "min_freq
)

display(viz_df)
```

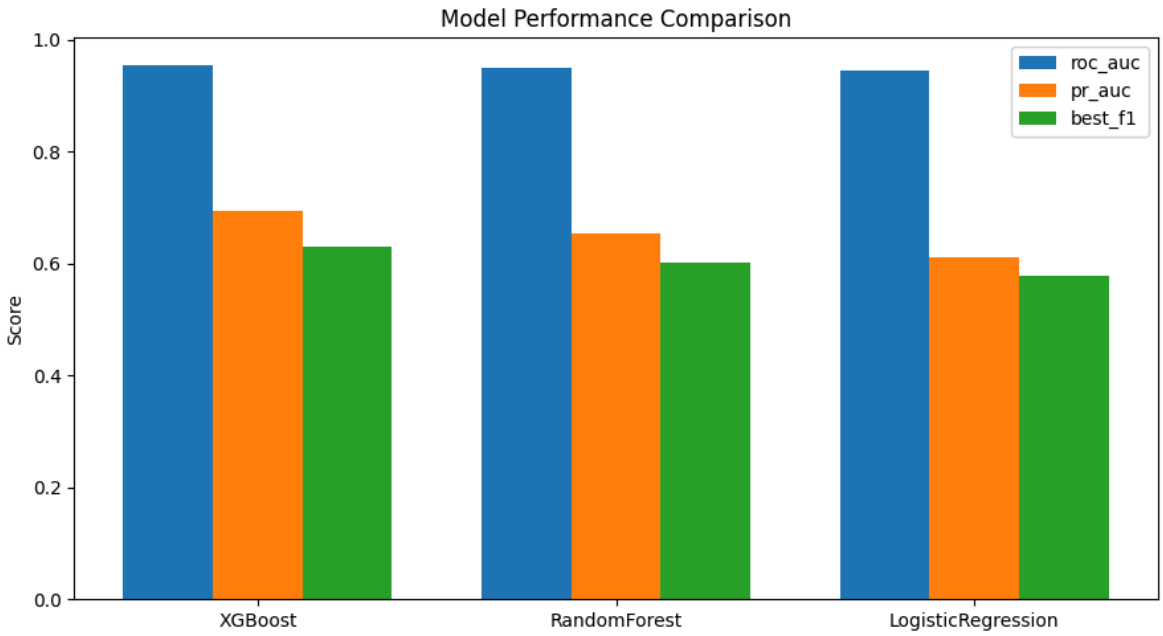
	model	roc_auc	pr_auc	best_threshold	best_f1	precision	recal
0	XGBoost	0.956083	0.695148	0.292965	0.630499	0.616275	0.645396
1	RandomForest	0.949795	0.654142	0.247540	0.602520	0.571325	0.637318
2	LogisticRegression	0.945792	0.612175	0.269160	0.579379	0.548538	0.613893
3	XGBoost	0.955962	0.694819	0.286896	0.633509	0.612599	0.655897
4	XGBoost	0.956083	0.695148	0.292965	0.630499	0.616275	0.645396

In [29]:

```
# This is a small helper block for turning the summary table back into figure in
# I kept it separate because it made the later plotting cells easier for me to t

# Select best row by F1
best_row = viz_df.sort_values("best_f1", ascending=False).iloc[0]

cm = np.array([
    [best_row["tn"], best_row["fp"]],
    [best_row["fn"], best_row["tp"]]
], dtype=int)
```

In [30]:

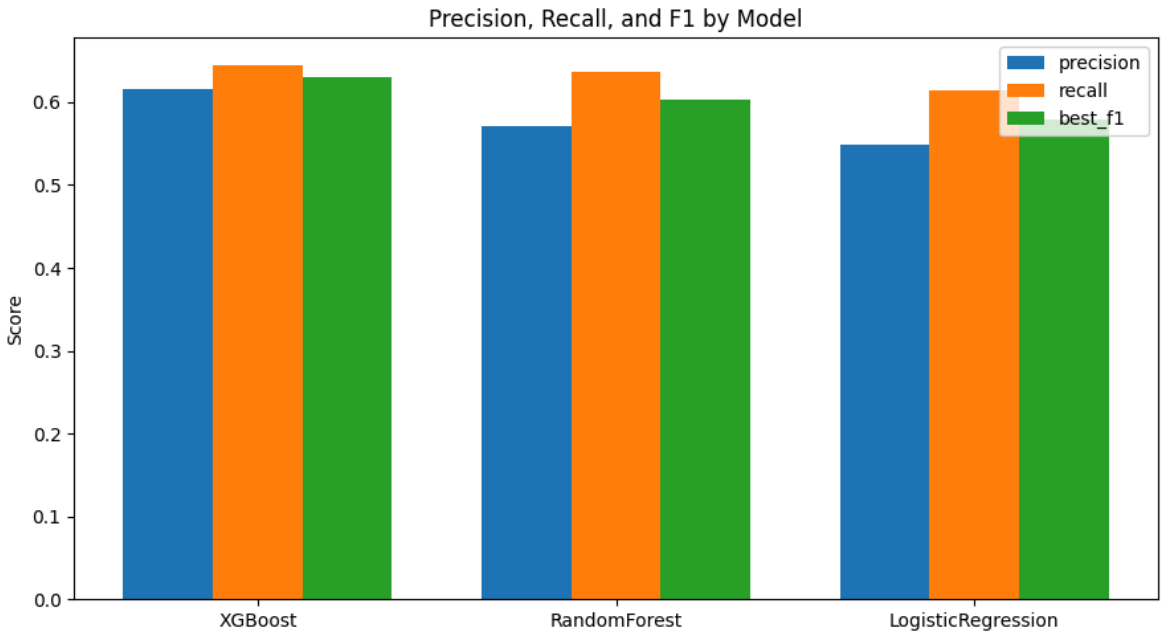
```
# From here on I am mostly experimenting with different report-style comparisons
# These cells are somewhat redundant with one another, but I am keeping them so
# still shows the path I took before settling on the final visuals.

model_comp_df = viz_df[viz_df["section"] == "Model comparison"].copy()
model_comp_df = model_comp_df.sort_values("best_f1", ascending=False)

metrics_to_plot = ["roc_auc", "pr_auc", "best_f1"]

x = np.arange(len(model_comp_df))
width = 0.25

fig, ax = plt.subplots(figsize=(9, 5))

for i, metric in enumerate(metrics_to_plot):
    ax.bar(x + (i - 1) * width, model_comp_df[metric], width=width, label=metric

ax.set_xticks(x)
ax.set_xticklabels(model_comp_df["model"], rotation=0)
ax.set_ylabel("Score")
ax.set_title("Model Performance Comparison")
ax.legend()

plt.tight_layout()
plt.show()

display(model_comp_df[["model", "roc_auc", "pr_auc", "best_f1", "precision", "re
```



	model	roc_auc	pr_auc	best_f1	precision	recall
0	XGBoost	0.956083	0.695148	0.630499	0.616275	0.645396
1	RandomForest	0.949795	0.654142	0.602520	0.571325	0.637318
2	LogisticRegression	0.945792	0.612175	0.579379	0.548538	0.613893

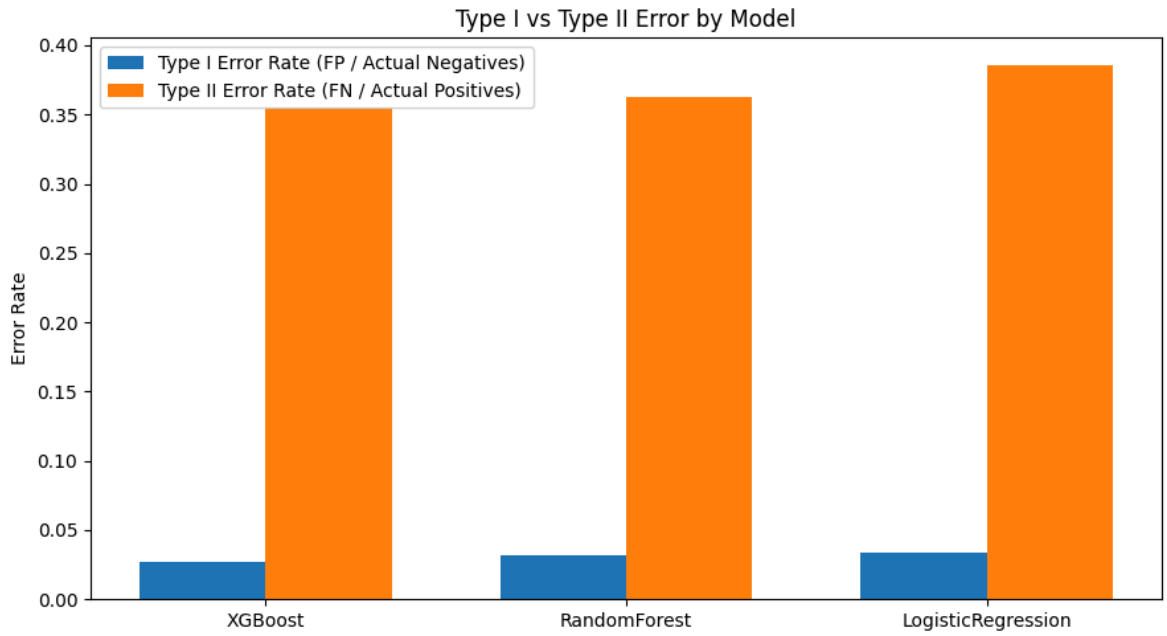
In [31]:

```
# This is another comparison view built from the same summary table.
# I did not end up needing every version, but I left them here because they help
# whether the ranking between models stayed stable across different metrics.

metrics_to_plot = ["precision", "recall", "best_f1"]

x = np.arange(len(model_comp_df))
width = 0.25

fig, ax = plt.subplots(figsize=(9, 5))

for i, metric in enumerate(metrics_to_plot):
    ax.bar(x + (i - 1) * width, model_comp_df[metric], width=width, label=metric

ax.set_xticks(x)
ax.set_xticklabels(model_comp_df["model"])
ax.set_ylabel("Score")
ax.set_title("Precision, Recall, and F1 by Model")
ax.legend()

plt.tight_layout()
plt.show()
```



In [32]:

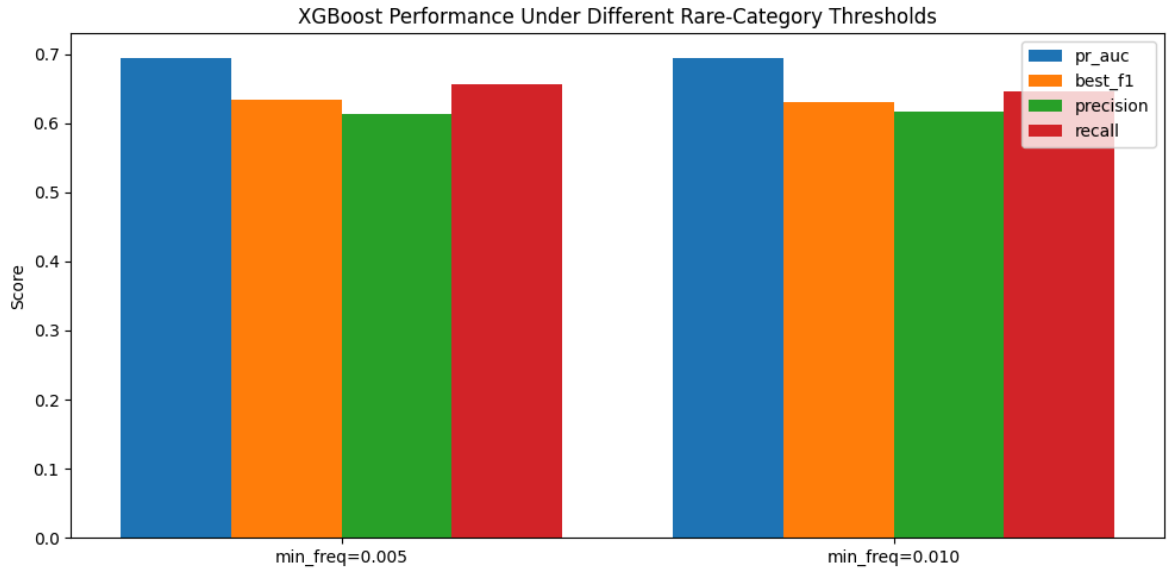
```
# Here I reshape the same model comparison results into error-style metrics.
# It overlaps with the earlier comparison plots, so I now treat it as optional s

model_comp_df = model_comp_df.copy()

model_comp_df["type1_error_rate"] = model_comp_df["fp"] / (model_comp_df["fp"] +
model_comp_df["type2_error_rate"] = model_comp_df["fn"] / (model_comp_df["fn"] +

x = np.arange(len(model_comp_df))
width = 0.35

fig, ax = plt.subplots(figsize=(9, 5))
ax.bar(x - width/2, model_comp_df["type1_error_rate"], width=width, label="Type
ax.bar(x + width/2, model_comp_df["type2_error_rate"], width=width, label="Type

ax.set_xticks(x)
ax.set_xticklabels(model_comp_df["model"])
ax.set_ylabel("Error Rate")
ax.set_title("Type I vs Type II Error by Model")
ax.legend()

plt.tight_layout()
plt.show()

display(model_comp_df[["model", "type1_error_rate", "type2_error_rate", "fp", "f
```



	model	type1_error_rate	type2_error_rate	fp	fn
0	XGBoost	0.026584	0.354604	995	878
1	RandomForest	0.031633	0.362682	1184	898
2	LogisticRegression	0.033423	0.386107	1251	956

In [33]:

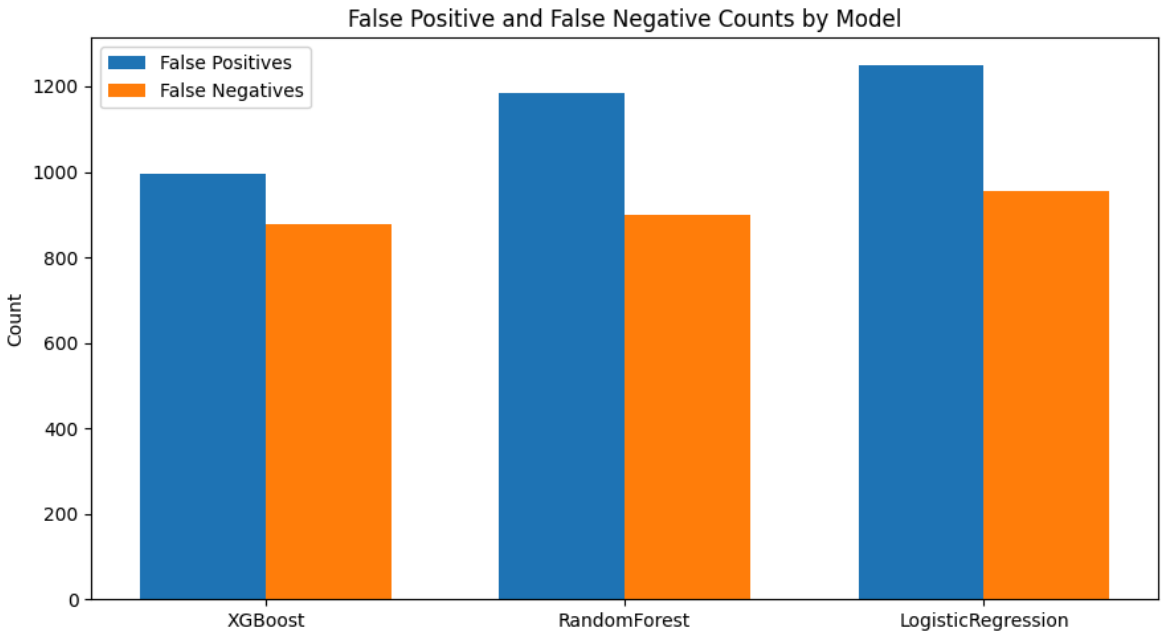
```
# This plot is more diagnostic than essential.
# I used it to sanity-check that my preprocessing choice was helping for the rig
# even though only a subset of these visuals made it into the final report.

xgb_freq_df = viz_df[viz_df["section"] == "XGBoost min_frequency test"].copy()
xgb_freq_df = xgb_freq_df.sort_values("min_frequency")

metrics_to_plot = ["pr_auc", "best_f1", "precision", "recall"]

x = np.arange(len(xgb_freq_df))
width = 0.2

fig, ax = plt.subplots(figsize=(10, 5))

for i, metric in enumerate(metrics_to_plot):
    ax.bar(x + (i - 1.5) * width, xgb_freq_df[metric], width=width, label=metric

ax.set_xticks(x)
ax.set_xticklabels([f"min_freq={mf:.3f}" for mf in xgb_freq_df["min_frequency"]]
ax.set_ylabel("Score")
ax.set_title("XGBoost Performance Under Different Rare-Category Thresholds")
ax.legend()

plt.tight_layout()
plt.show()

display(xgb_freq_df[["display_name", "roc_auc", "pr_auc", "best_f1", "precision"
```



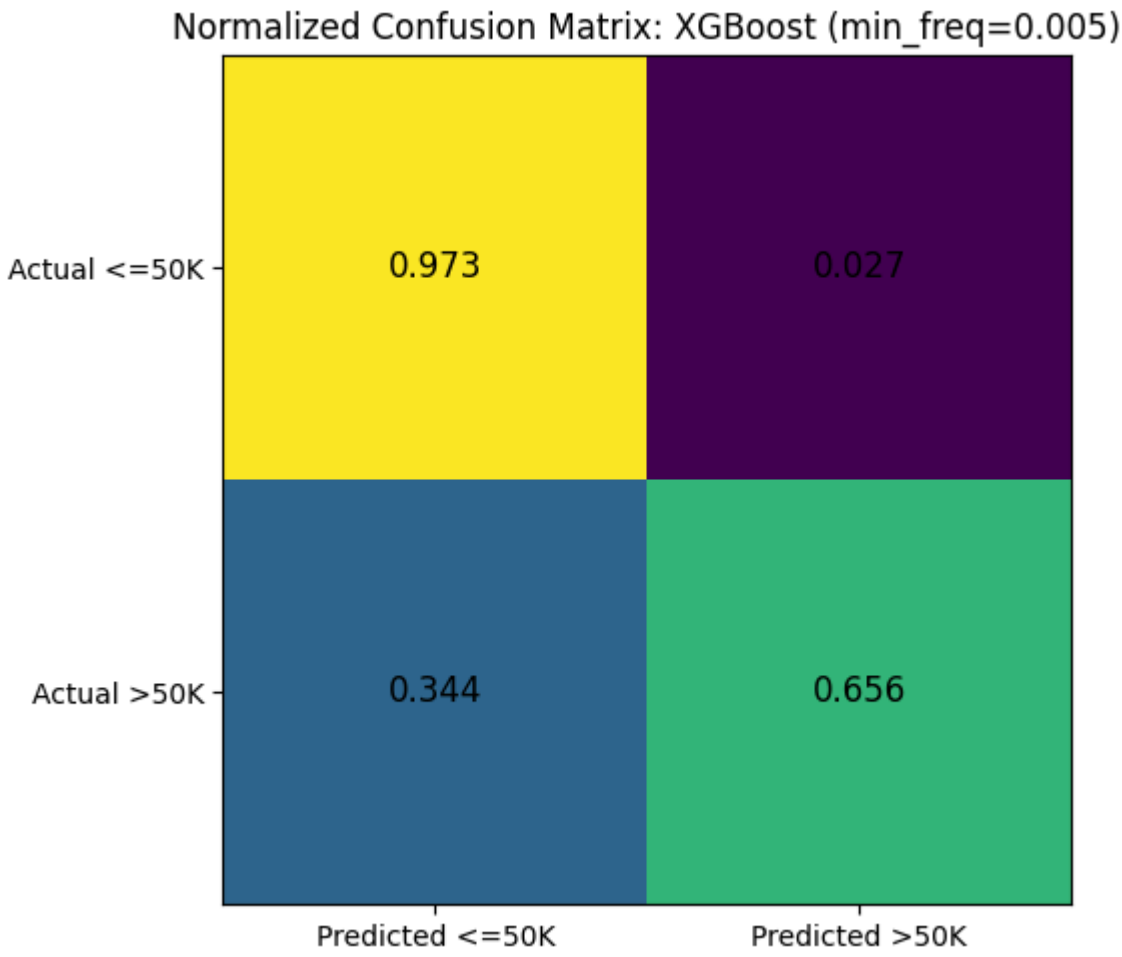
	display_name	roc_auc	pr_auc	best_f1	precision	recall
3	XGBoost (min_freq=0.005)	0.955962	0.694819	0.633509	0.612599	0.655897
4	XGBoost (min_freq=0.01)	0.956083	0.695148	0.630499	0.616275	0.645396

In [34]:

```
# This is another draft view for the model comparison section.
# It is somewhat repetitive, but I am leaving it in place because it reflects ho
# before deciding which figure was the cleanest to keep.

fig, ax = plt.subplots(figsize=(9, 5))

x = np.arange(len(model_comp_df))
width = 0.35

ax.bar(x - width/2, model_comp_df["fp"], width=width, label="False Positives")
ax.bar(x + width/2, model_comp_df["fn"], width=width, label="False Negatives")

ax.set_xticks(x)
ax.set_xticklabels(model_comp_df["model"])
ax.set_ylabel("Count")
ax.set_title("False Positive and False Negative Counts by Model")
ax.legend()

plt.tight_layout()
plt.show()
```

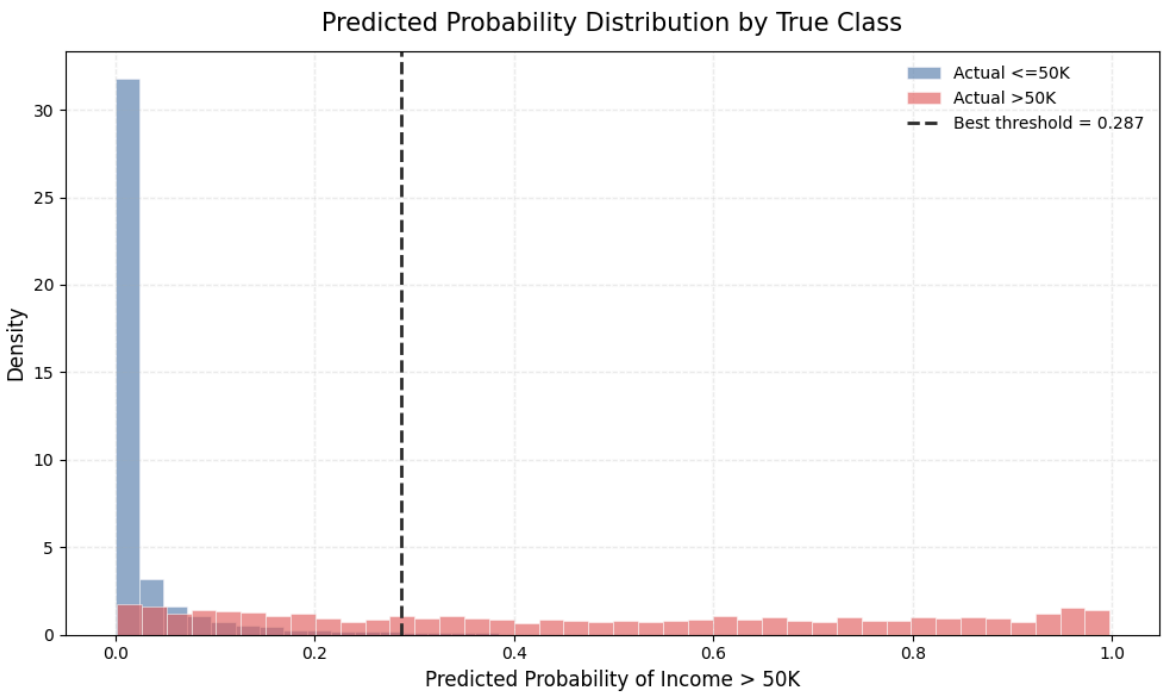


In [35]:
```
# This is another reporting cell that overlaps with earlier plotting work.
# I am leaving it here because it was part of how I iterated on the confusion-ma

cm_norm = cm / cm.sum(axis=1, keepdims=True)

fig, ax = plt.subplots(figsize=(6, 5))
im = ax.imshow(cm_norm)

ax.set_xticks([0, 1])
ax.set_yticks([0, 1])
ax.set_xticklabels(["Predicted <=50K", "Predicted >50K"])
ax.set_yticklabels(["Actual <=50K", "Actual >50K"])
ax.set_title(f"Normalized Confusion Matrix: {best_row['display_name']}")

for i in range(2):
    for j in range(2):
        ax.text(j, i, f"{cm_norm[i, j]:.3f}", ha="center", va="center", fontsize

plt.tight_layout()
plt.show()
```



In [36]:
```
# I kept this section in notebook-style scratch format because I originally used
# to test report figures quickly. It overlaps with earlier cells, so I now treat
# optional supporting work rather than the main analysis path.

# %%
# ============================================================
# PREDICTED PROBABILITY DISTRIBUTION
# ============================================================

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# 假設你已經有：
# y_test
# final_results["y_prob"]
# final_results["best_threshold"]

y_prob_final = final_results["y_prob"]
best_threshold = final_results["best_threshold"]

prob_df = pd.DataFrame({
    "y_true": np.asarray(y_test),
    "y_prob": np.asarray(y_prob_final)
})

neg_probs = prob_df.loc[prob_df["y_true"] == 0, "y_prob"]
pos_probs = prob_df.loc[prob_df["y_true"] == 1, "y_prob"]

plt.figure(figsize=(10, 6))
```

```
# Professional colors
color_neg = "#4C78A8"   # muted blue
color_pos = "#E45756"   # muted red
threshold_color = "#2F2F2F"

plt.hist(
    neg_probs,
    bins=40,
    density=True,
    alpha=0.60,
    label="Actual <=50K",
    color=color_neg,
    edgecolor="white",
    linewidth=0.6
)

plt.hist(
    pos_probs,
    bins=40,
    density=True,
    alpha=0.60,
    label="Actual >50K",
    color=color_pos,
    edgecolor="white",
    linewidth=0.6
)

plt.axvline(
    best_threshold,
    color=threshold_color,
    linestyle="--",
    linewidth=2,
    label=f"Best threshold = {best_threshold:.3f}"
)

plt.title("Predicted Probability Distribution by True Class", fontsize=15, pad=1
plt.xlabel("Predicted Probability of Income > 50K", fontsize=12)
plt.ylabel("Density", fontsize=12)
plt.legend(frameon=False)
plt.grid(alpha=0.2, linestyle="--")
plt.tight_layout()
plt.show()
```



In [37]:
```
final_results
```

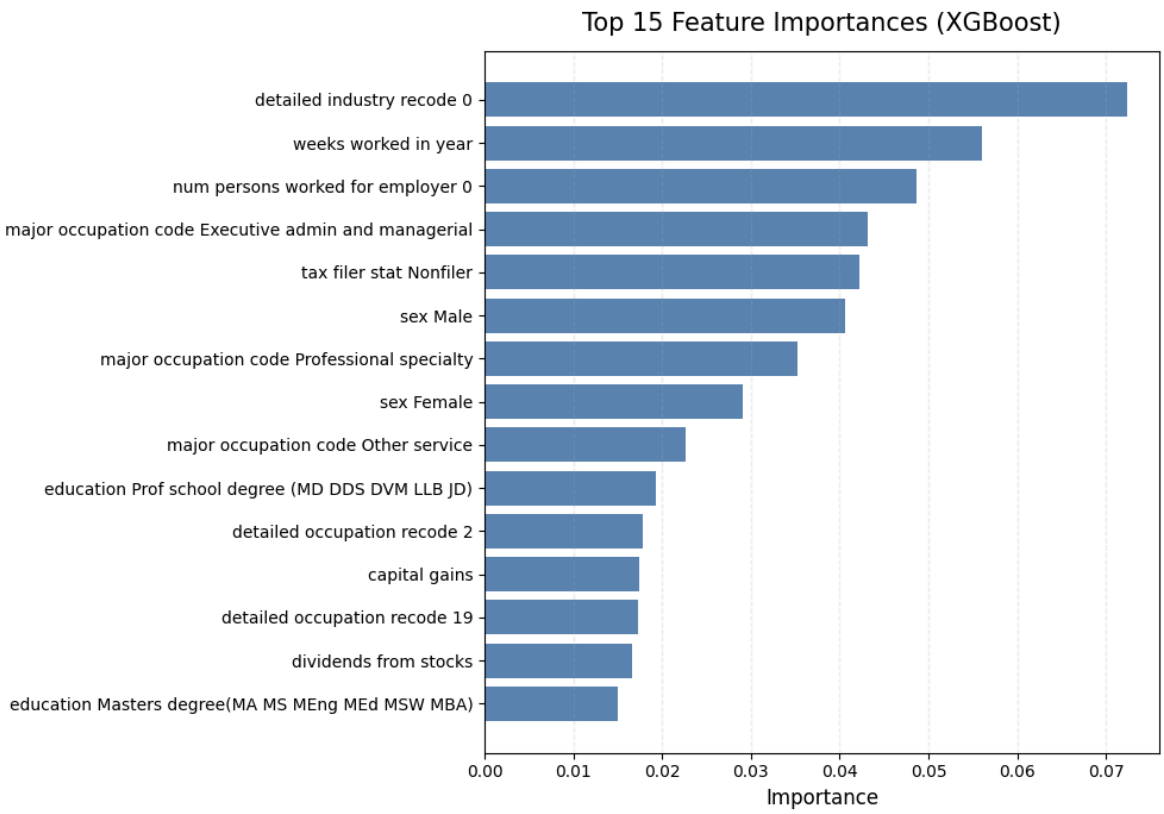
Out[37]:
```
{'roc_auc': 0.9559618715473402,
 'pr_auc': 0.694818829892732,
 'best_threshold': np.float32(0.28689602),
 'best_f1': np.float64(0.6335088740861101),
 'precision': np.float64(0.6125990192380234),
 'recall': np.float64(0.6558966074313409),
 'tn': np.int64(36402),
 'fp': np.int64(1027),
 'fn': np.int64(852),
 'tp': np.int64(1624),
 'y_prob': array([5.9758375e-05, 3.3691670e-03, 1.0808577e-03, ..., 1.0520163e-
02,
        1.4930086e-03, 9.3017705e-03], shape=(39905,), dtype=float32),
 'y_pred': array([0, 0, 0, ..., 0, 0, 0], shape=(39905,)),
 'confusion_matrix': array([[36402,  1027],
        [  852,  1624]])}
```

In [38]:
```
# This was another scratch-area version of the reporting workflow.
# I did not delete it because it still documents part of the path I took while r

# # %%
# # ============================================================
# # TOP FEATURE IMPORTANCE
# # ============================================================

# import numpy as np
# import pandas as pd
# import matplotlib.pyplot as plt

preprocessor = final_pipe.named_steps["preprocessor"]
final_model = final_pipe.named_steps["model"]

feature_names = preprocessor.get_feature_names_out()
importances = final_model.feature_importances_

importance_df = pd.DataFrame({
```

```
    "feature": feature_names,
    "importance": importances
})

importance_df = importance_df.sort_values("importance", ascending=False).head(15
importance_df = importance_df.sort_values("importance", ascending=True)

# plt.figure(figsize=(10, 7))

# bar_color = "#4C78A8"

# plt.barh(
#     importance_df["feature"],
#     importance_df["importance"],
#     color=bar_color,
#     alpha=0.9
# )

# plt.title("Top 15 Feature Importances (XGBoost)", fontsize=15, pad=12)
# plt.xlabel("Importance", fontsize=12)
# plt.ylabel("")
# plt.grid(axis="x", alpha=0.2, linestyle="--")
# plt.tight_layout()
# plt.show()
```

In [39]:

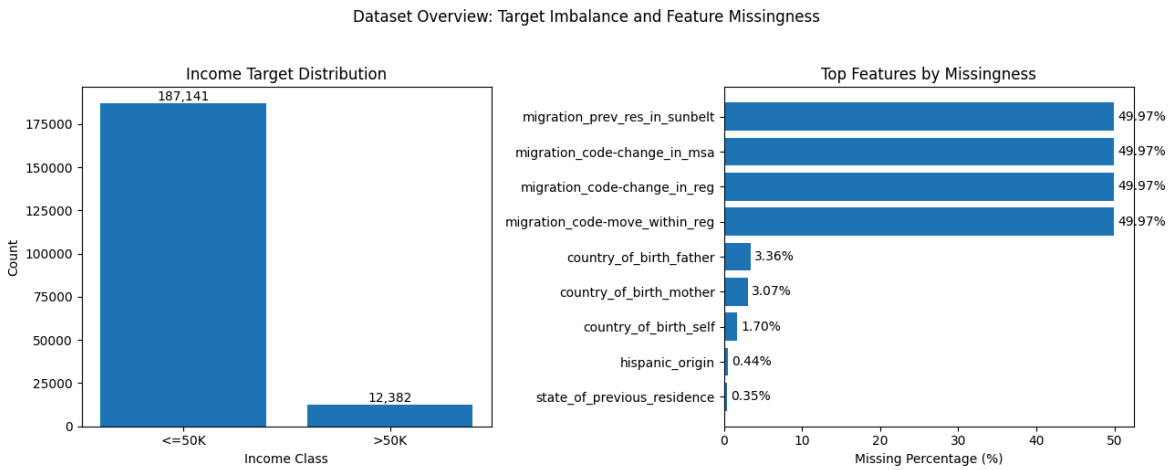
```
# This block is also mostly report-polishing work rather than new analysis.
# It overlaps with earlier figure-generation code, so I would treat it as option
# the notebook further for a final handoff.

# %%
# ============================================================
# CLEAN FEATURE NAMES FOR DISPLAY
# ============================================================

def clean_feature_name(name):
    name = name.replace("num__", "")
    name = name.replace("cat__", "")
    name = name.replace("remainder__", "")
    name = name.replace("_", " ")
    return name

importance_df["feature"] = importance_df["feature"].apply(clean_feature_name)

plt.figure(figsize=(10, 7))
plt.barh(
    importance_df["feature"],
    importance_df["importance"],
    color="#4C78A8",
    alpha=0.9
)

plt.title("Top 15 Feature Importances (XGBoost)", fontsize=15, pad=12)
plt.xlabel("Importance", fontsize=12)
plt.ylabel("")
plt.grid(axis="x", alpha=0.2, linestyle="--")
plt.tight_layout()
plt.show()
```



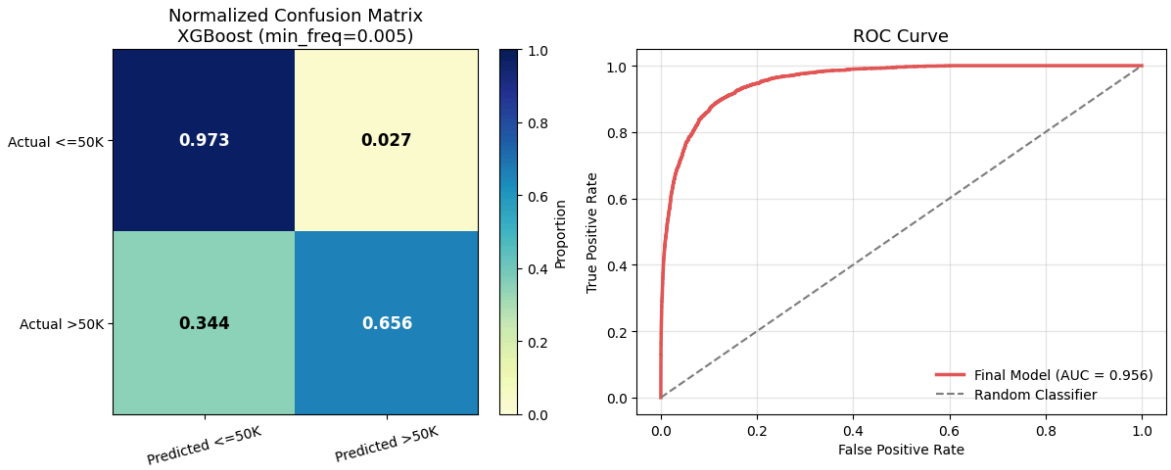
In [40]:

```
# This block is mainly for an alternate report figure.
# It is somewhat redundant with the earlier plotting section, but I kept it beca
# one of the intermediate versions I used while refining the final deliverables.

import pandas as pd
import matplotlib.pyplot as plt

# Assume df_model already exists
# If not, replace df_model with your cleaned dataframe name, e.g. df

plot_df = df_model.copy()

# -------- Panel 1: target distribution --------
target_counts = plot_df["target"].value_counts().sort_index()

# -------- Panel 2: top missing features --------
missing_pct = (
    plot_df.isna().mean()
    .sort_values(ascending=False)
)

missing_pct = missing_pct[missing_pct > 0].head(10)

# -------- Combined figure --------
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Left: target distribution
axes[0].bar(
    ["<=50K", ">50K"],
    target_counts.values
)
axes[0].set_title("Income Target Distribution")
axes[0].set_ylabel("Count")
axes[0].set_xlabel("Income Class")
```

```
for i, v in enumerate(target_counts.values):
    axes[0].text(i, v, f"{v:,}", ha="center", va="bottom")

# Right: missingness
axes[1].barh(
    missing_pct.index[::-1],
    missing_pct.values[::-1] * 100
)
axes[1].set_title("Top Features by Missingness")
axes[1].set_xlabel("Missing Percentage (%)")

for i, v in enumerate((missing_pct.values[::-1] * 100)):
    axes[1].text(v + 0.5, i, f"{v:.2f}%", va="center")

plt.suptitle("Dataset Overview: Target Imbalance and Feature Missingness", y=1.0
plt.tight_layout()
plt.show()
```



In [41]:

```
# This final figure cell overlaps with earlier ROC / confusion-matrix code.
# I kept it instead of deleting it because it shows the version I used when poli

from sklearn.metrics import roc_curve
import matplotlib.pyplot as plt
import numpy as np

# Normalize confusion matrix
cm_norm = cm / cm.sum(axis=1, keepdims=True)

# ROC values
fpr, tpr, _ = roc_curve(y_test, y_prob_final)

# Create side-by-side plots
fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# ============================================================
# Left: Normalized Confusion Matrix
# ============================================================
ax = axes[0]
im = ax.imshow(cm_norm, cmap="YlGnBu", vmin=0, vmax=1)

ax.set_xticks([0, 1])
ax.set_yticks([0, 1])
ax.set_xticklabels(["Predicted <=50K", "Predicted >50K"], rotation=15)
ax.set_yticklabels(["Actual <=50K", "Actual >50K"])
ax.set_title(f"Normalized Confusion Matrix\n{best_row['display_name']}", fontsiz
```

```python
for i in range(2):
    for j in range(2):
        val = cm_norm[i, j]
        ax.text(
            j, i, f"{val:.3f}",
            ha="center", va="center",
            fontsize=12,
            color="white" if val > 0.5 else "black",
            fontweight="bold"
        )

cbar = fig.colorbar(im, ax=ax, fraction=0.046, pad=0.04)
cbar.ax.set_ylabel("Proportion", rotation=90)

# ============================================================
# Right: ROC Curve
# ============================================================
ax2 = axes[1]
ax2.plot(fpr, tpr, linewidth=2.5, label=f"Final Model (AUC = {final_results['roc
ax2.plot([0, 1], [0, 1], linestyle="--", linewidth=1.5, color="gray", label="Ran

ax2.set_xlabel("False Positive Rate")
ax2.set_ylabel("True Positive Rate")
ax2.set_title("ROC Curve", fontsize=13)
ax2.legend(frameon=False)
ax2.grid(alpha=0.3)

plt.tight_layout()
plt.show()
```